

4. 🏆 **Team 18 won the JPMorgan Chase Code for Good 2025 Hackathon** held in Mumbai on June 7–8! Among the standout contributors was **Hritajit Sur** (3rd from left in Image 1), a brilliant student from our **IEDC Lab, UEM Kolkata**, who helped build **FootPathshala** — a smart, field-ready solution for UPAY that automates attendance, assessments, curriculum, and volunteer tracking. Kudos to JPMorgan Chase for this inspiring platform and to UEM for nurturing future-ready talent!



Industry-Academic Collaboration

1. BIRAC-Funded Project

Collaborating institutions: Regional Institute of Ophthalmology (RIO), Indian Institute of Science (IISc), UEM

🔍 **Leads:** Apurba Nandi, Avik Kumar Das

📄 **Output:** Copyrighted OCT dataset, AI models for diabetic retinopathy diagnosis

2. Research Collaboration with Jadavpur University

📄 **Outcome:** Publication in *Journal of Cognitive Neurodynamics* (🧠 Impact Factor: 3.10)

3. Ongoing Project with IEST, Shibpur

🔍 **Focus Area:** Research on underwater technologies

📍 **Outcome:** Active interdisciplinary research in progress

4. Collaboration with WEBEL

🏆 **Outcome:** Industrial training provided to lab students, strengthening industry-readiness and practical exposure

Product Development Highlights

- 📄 **2 Copyrights Filed:**
 - OCT Image Dataset – For retinal disease detection (BIRAC collaboration)
 - Temple Architecture Dataset – For cultural heritage preservation and classification
- 🔒 **12 Patents Filed in Diverse Domains:**
 - Agriculture – Smart irrigation, disease detection
 - Medical Diagnostics – AI-based early disease screening

- Electrical & Industrial Automation – Smart control systems
- Forest Fire Sustainability – Real-time wildfire monitoring
- Smart Traffic Management – AI-driven congestion and violation detection

Research Outcomes Summary

RESEARCH OUTCOMES	ACCEPTED/ PUBLISHED	COMMUNICATED
<i>Copyrights</i>	-	2
<i>Patents</i>	-	12
<i>Transactions</i>	-	2
<i>Journals</i>	2	8
<i>Conferences</i>	27	30
<i>Book Chapters</i>	9	10
<i>Awards & Recognitions</i>	3	NA

IEDC Internship Highlights

At our IEDC Lab, we successfully organized two **academic internship programs**, engaging over **200 students** in hands-on learning experiences. These programs provided practical exposure to **prototype designing, AI model development, and real-world problem solving**, equipping students with the skills needed to meet industry demands. Through guided mentorship and project-based learning, participants gained valuable insights into cutting-edge technologies and innovation-driven practices. The initiative has empowered students to transform their ideas into tangible solutions, making them industry-ready and future-focused

♦ **Major Research Grant Received**

We are delighted to share that we have received a generous research grant of approximately ₹50 lakhs from the IEM-UEM Group Trust and other supporting bodies. This significant support will further strengthen our ongoing research and innovation initiatives. We extend our heartfelt gratitude to **Prof. (Dr.) Satyajit Chakrabarti** Sir for his visionary leadership and unwavering commitment to advancing research and development. His initiative has opened new avenues for impactful work and collaboration.
